



هيئة المساحة الجيولوجية السعودية
Saudi Geological Survey

Qualification Request Advanced analytics platform for the Geological Data Center in the Kingdom of Saudi Arabia

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*Request for Qualification of the Geological Data Analysis Center
Advanced Analytics Platform*



Call for Pre-Qualification

The Saudi Geological Survey (SGS) intends to implement a new Advanced Analytics Platform project as a first step in the development of the new Saudi Center for Geoscience Data Analytics (GDAC-SA), which is prominent among the objectives of Saudi Arabia's Vision 2030 as it seeks to diversify the Kingdom's economy through the mineral resources sector. The Saudi Geological Survey (SGS) is pleased to invite you to participate in the pre-qualification process for **the Advanced Analytics Platform project of the Saudi Center for Geological Science Data Analysis (GDAC-SA).**

The purpose of this RFQ is to assess the capabilities of potential bidders for the Saudi Geoscience Data Analytics Center's AI platform. Qualified bidders will then be invited to move from the RFP stage to the RFP stage. Due to the complex nature of the RFP/RFP process, bidders can apply in consortia to meet all the requirements.

The SGS invites you to submit Request for Qualifications (RFQ) to complete the qualification requirements and participate in the bidding process for this project. In order for the application to be considered, the application must meet all the required elements contained in this qualification application form and submit all required documents and certificates, as any lack of documentation may result in the disqualification of the application.

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Section One: Introduction

Introduction:

In April 2016, the Kingdom of Saudi Arabia officially announced the "Saudi Vision 2030" (hereinafter referred to as "Vision 2030") and the "National Transformation Plan", which aspires to diversify the Saudi economy, with digitalization and artificial intelligence as the most important elements of this transformation. As Saudi Arabia strives to be among the top 15 countries in AI, among the top 10 countries with open-source data, and among the top 20 contributors to research and innovation globally by 2030, Saudi Arabia's development goals are to achieve its economic transformation, in part, through the further development of the big data and artificial intelligence industries.

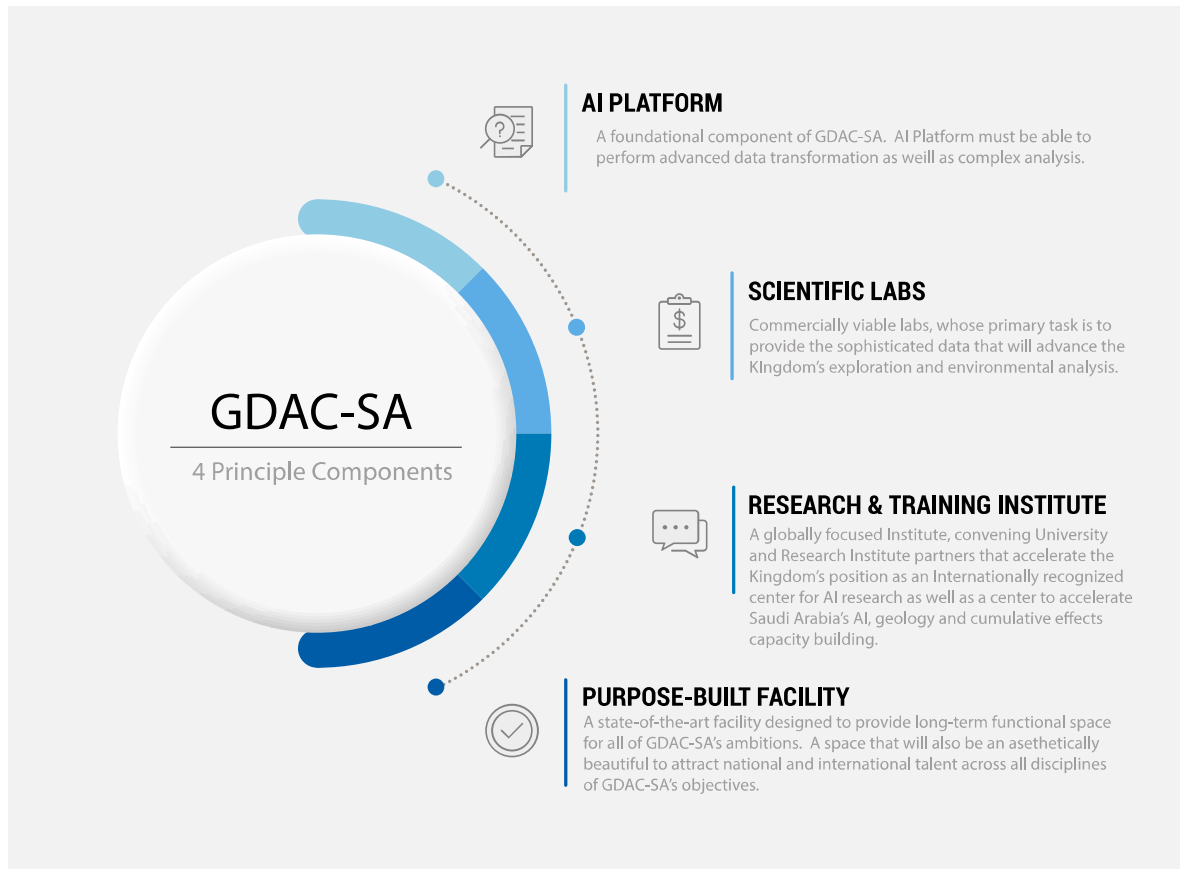
The Saudi Geological Survey is one of the entities of the Ministry of Industry and Mineral Resources, which is the main entity for the creation and storage of geological data and providing access to it in the Kingdom, as the Kingdom of Saudi Arabia in general has not yet been sufficiently explored in terms of solid minerals, and there are many untapped opportunities in the mining sector in the Kingdom of Saudi Arabia, as the Arabian Shield area located within the Kingdom of Saudi Arabia, which extends over an area of 600,000 square kilometers, is considered a major source of precious and essential minerals. More than 90 species and at least 15 of them are commercially viable.



Shape 1: Critical and strategic minerals

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The Saudi Geological Survey has received approval to establish a Geological Data Analysis Center that will be powered by an AI/Big Data Analytics Platform. The platform will enable the transformation of data and analytics related to geological exploration as well as the environmental and social impacts associated with mining, and the Geological Data Analysis Center will also develop geoscientific laboratories to produce the data required for the analytical requirements of the center. A new research and training institute will be established to be a catalyst for pioneering research and building capacity within the Kingdom, and a state-of-the-art facility dedicated to this purpose will be built on the Geological Survey campus.

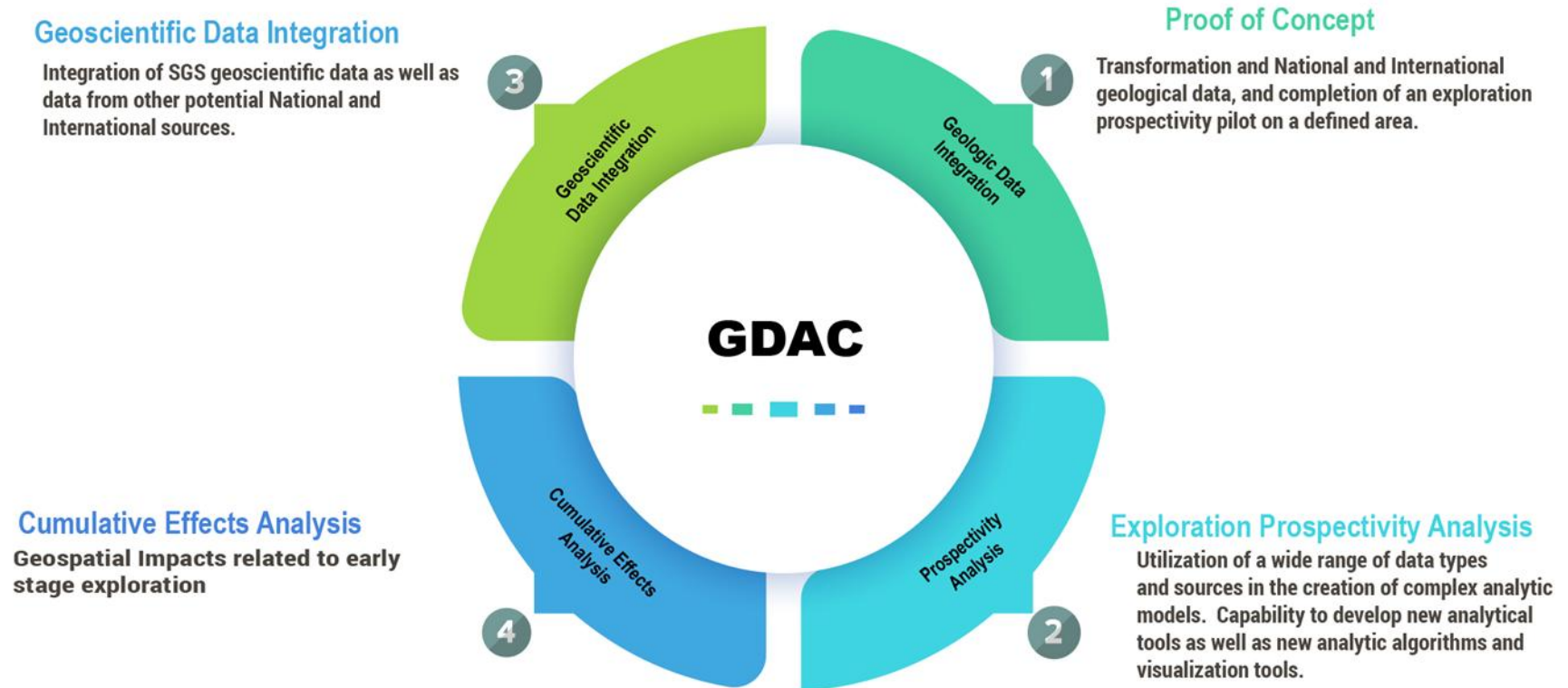


Shape 2: The four components of the platform

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Vision of the Geological Data Analysis Center Platform

The Geological Data Analysis Center should primarily support mineral exploration as a core objective of Vision 2030, and mineral exploration will be considered as the main purpose and product of the Center. As shown in Figure 2, the AI platform must be able to grow and evolve to keep pace with the broad research interests of the Geological Data Analysis Center in order to advance the research and AI goals in Saudi Arabia.



Shape 3: Vision of the Saudi Center for Geological Data Analysis Platform

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The AI platform must be able to perform data transformations related to exploration geology and its analysis, assess the environmental and social impact of mining, and the analytics platform provider must be able to deliver and enable world-class AI and analytics applications. The first and second phases of the development of the Geological Data Analysis Center and its capabilities will be related to the integration of geological data, mineral exploration and prospecting, as the identification of high-potential mineral resources that are often hidden and extremely deep makes the field of exploration a highly complex science because of its diverse methods and data, including reports, geochemistry, geophysics, remote sensing, geology, drilling records, etc., in addition to many analytical features, such as the possibility of the presence of minerals, rocks, and structures. and variable metals, and others. Each type of data and every geological feature helps tell a possible story of what the Earth is hiding. However, the primary goal of mineral exploration analytics is the ability to aggregate this data, normalize it and model multiple important features as they are likely to be productive from a given geographic location.

The first phase: the proof of concept phase

To achieve the goals of the Geological Data Analysis Center for Mineral Exploration, the AI platform will be developed by integrating a wide range of data types, creating engineering features from different forms of data. Most of this data will be at the national and regional levels and contain aggregations of data and gaps, including structured and unstructured data, some of which are without metadata including the locations of this data which is mostly obtained from the National Geological Database (NGD) and the National Information Center (NIC). The phase will require the Proponent also has the ability to integrate complex features from different types of data – such as natural language processing, image analysis, and/or the development of new algorithms. A large amount of new geological data is currently being collected, such as detailed geological maps and geochemistry, which will be part of the required data conversion. The Proponent that is subsequently awarded the scope of a specific area within the Kingdom to implement an initial pilot project for exploratory forecasting. Upon successful completion of the

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proof-of-concept model, subsequent phases begin. The winner will also be given a period not exceeding one year to implement the proof-of-concept model.

The AI platform should be able to integrate new data from internal and external sources within the first phase and subsequent phases as the first phase is essentially aimed at creating the underlying data ecosystem that should exist in a single geospatial analytics platform, and this will require proponents to create a data structure based on the core features associated with geology. The data warehouse then fills this underlying data using a combination of techniques such as natural language processing and feature engineering, which supports the operation of the Geological Data Analysis Center. The proponent that is subsequently awarded the scope of a specific area within the Kingdom to implement an initial pilot project for exploratory forecasting. Upon successful completion of the proof-of-concept model, subsequent phases begin. The winner will also be given a period not exceeding one year to implement the proof-of-concept model.

The Second Phase of the Artificial Intelligence Platform at the Saudi Center for Geological Data Analysis: Exploration Prospects Phase

The second step is designed to support the Saudi Geological Survey (SGS) and the Geological Data Analysis Center (GDAC) by providing them with advanced AI-based tools to enhance their capabilities to perform advanced geological analyses at the level of resources, commodities or sediment types. To achieve this goal, the platform must include a range of ready-made analytical methods, as well as capabilities that allow for the rapid development of new analysis methods, as well as the ability to build advanced taxonomic models that mimic mineral systems. It identifies important characteristics for the discovery of new deposits.

Phase III of the Artificial Intelligence Platform of the Saudi Center for Geological Data Analysis: Integration of Geoscience Data

As part of the vision of the Geological Data Analysis Center, the Saudi Geological Survey aims to create an AI platform that can continue to grow and expand to cover other geoscience-related implications, and as in Phase 1, Phase 3 will focus on creating a more expansive data

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integration environment. Starting with other geoscience-related topics, the GDAC's AI platform will integrate the data into one unified data structure – all based on geospatial and the tools and techniques used in the first phase, and integrate other secondary analytical topics and industry categories as described above.

Phase IV: Cumulative Impact Analysis

Like most complex analytical challenges, changes to the natural environment from various factors have intentional and unintended effects on human, environmental, and economic systems, and these effects rarely come from just one source, and it is often other factors such as time, duration, concentration, and a combination of factors that cause the effects on complex systems. Cumulative effects research typically involves monitoring and analyzing multiple variables over time to assess how different factors interact and the changes they bring about in the ecosystem under study. This can include studying changes in the built environment, biodiversity, ecosystem function, water quality and other indicators of ecosystem health. By understanding how different factors interact and accumulate over time, cumulative impact research can help make decisions about how to manage and mitigate the impact of human activities on the environment. More detailed information will be provided during the RFP process.

Integration of Stages 1-4

Enabling Saudi Arabia's strategy to develop the mining sector as a third economic pillar is a strategic necessity centered on the capabilities of the Geological Data Analysis Center, particularly through its AI platform. The four components of the development of the AI platform of the Saudi Center for Geoscience Data Analysis are formed through four primary objectives: developing the analytical model by type of commodity or mineral resources, generating new targets, developing extracted resources and environmental assessment of high-potential resources of them.

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Competition Dates

The deadline for receiving the qualification application proposals is set in the (Etimad) platform.

Required Documents and Licenses

The following certificates and documents must be submitted numbered and attested with the company's seal. Official and legal documents include:

1. Valid Commercial Registration Certificate
2. Valid Zakat and Income Certificate as well as VAT Certificate if registered.
3. Valid certificate from the General Organization for Social Insurance.
4. Valid Chamber of Commerce Certificate.
5. A valid investment license if the competitor is licensed under the Foreign Investment Regulation.
(Foreign bidders must be registered with the Ministry of Investment of the Kingdom of Saudi Arabia.)
6. Valid certificate of achievement of the Emiratization rate of Saudi cadres (Saudization / Taqat certificate).
7. Valid quality policy/certificate and HSE policy/certificate.
8. When applying for an alliance, a copy of the alliance agreement must be submitted.
9. The approved financial statements of the entity's budget during the past three years.
10. Company information and required registration certificates: Yes/No

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Government Entity Representative

Technical inquiries regarding this qualification application form should be sent in writing to:

Dr. Wadee Kashghari, Owner of the Regional Geological Program initiative

Office Phone: +966-2 6195000 ext. 5222

Email: TI-RGP@sgs.gov.sa

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Section Two: General Conditions

The purpose of the Request for Qualifications (RFQ) is to provide the opportunity for companies or consortiums with experience and qualification to implement the Advanced Analytics Platform project of the Saudi Center for Geological Science Data Analysis. Eligible companies are invited to submit their bids (financial and technical) for the Saudi Center for Geoscience Data Analysis (GDAC-SA) Advanced Analytics Platform briefly described here based on the Proposal and Specification Booklet (RFP), which will be sent to the qualified bidders.

The bidder can be either a single company/entity or a group of interested and qualified entities. In the case of an alliance, qualification applications must be submitted to all members of that alliance, including all partners and subcontractors. To be considered for the qualification process, the bidder must provide the following:

1. A brief statement of the company/entity, its history and any related projects implemented in the last five years.
2. Formal agreements between trade alliance partners as well as subcontractors.

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Section Three: Presentation

How to submit proposals

The offer must be submitted in a single batch of documents.

Application Instructions

Please read and follow the following instructions: Any violation of any of the following instructions will result in disqualification. The application shall be submitted to the Saudi Geological Survey in a correct form by an authorized person as follows:

1. The application must be submitted in advance in either Arabic and/or English and in accordance with the requirements set out in this qualification application form. All documents must be in a searchable format.
2. One contact administrator shall be identified, stating their job title and all contact details.
3. The presentation should contain concise written materials and illustrations to allow for a deeper understanding and clearer assessment of the applicant's abilities. Only the information required in the qualification application will be provided, and additional information that is not required will be discarded when included in the application.
4. A complete, searchable structured digital copy of the qualification applications must be submitted via the platform before the deadline for submitting the pre-qualification application.
5. Applications will not be received after the deadline, and the applicant will be disqualified.
6. All attachments and forms must be filled out electronically.
7. All copies of official documents must be stamped with the applicant's seal to prove the accuracy of the information received.
8. If the attached forms are not filled out with the correct information, the applicant will be disqualified.
9. Failure to provide the necessary data and documents in full, or failure to fill in the

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required data in the mentioned annexes, will result in disqualification from the competition.

10. The results of the qualification will be announced on the tenders and procurement platform, and if this is not possible, it will be announced on the official website of the Geological Survey.
11. Excluded competitors can inquire about the reasons for disqualification by contacting the entity's representative.
12. All documents submitted are the property of the Geological Survey and will not be returned to the bidders once submitted.

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Section Four: Evaluation of Proposals

Criteria for evaluating proposals

All applications for qualification are evaluated electronically through the Etimad platform, and the total score for the evaluation of the qualification application must exceed 70% of the total to be eligible to receive the RFP and participate in the bidding process. Technical and administrative capacities constitute 60 per cent, and financial capacities 40 per cent of the total grade. The Geological Survey reserves the right to use evaluation criteria (whether subjective or objective) based on the Government Tenders and Procurement System.

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Table 1a: Technical and administrative criteria for evaluation.

Conclusion	Technical and Administrative Criteria for Evaluation	Number
Yes/No	Company Information and Required Registration Certificates	1
%٤٠	Previous Experience	2
%١٠	Quality Assurance	3
%١٠	Ensuring Environmental, Health and Safety	4
%٢٠	Existing contractual obligations	5
20%	Human Resources	6
100%	Total	

Figure 6: Technical and Management Evaluation Criteria

Table 1b: Financial evaluation criteria

Grade	Financial Evaluation Criteria	Number
%٤٠	Cash Ratio (Cash & Cash (Equivalents/Working Liabilities	1
%٣٠	Turnover Ratio (Current Assets/Current (Liabilities	2
30%	Quick Liquidity Ratio (Cash & Cash / (Equivalents + Accounts Receivable Current Liabilities	3
100%	Total	

Figure 7: Financial Evaluation Criteria

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Review and Selection

The Geological Survey will notify the successful applicants for the qualification assessment, and the results of the evaluation of the successful bidders will be published on the platform.

Geological Survey General Program Rights

Failure to comply with any of the requirements of this qualification application will result in the disqualification of the applicant. Notwithstanding anything else in this RFP, the Geological Survey has the right to change the dates, attachments, deadlines, and requirements described in this RFP, including the exclusion of applicants or the cancellation of the RFP process entirely.

Confidentiality and Conflicts of Interest

By submitting this qualification application, the applicant undertakes to maintain the confidentiality of any information or data obtained during the qualification process and not to disclose any such information to any third party during the application period, without obtaining written permission from the Geological Survey.

The applicant must promptly disclose in writing any potential conflict of interest to the Geological Survey, which will consider the nature of the applicant's responsibilities and the degree of the conflict that was likely or manifest and determine the manner that the applicant should take to address such conflict of interest.

Compliance with laws

The Winning Bidder agrees to comply with and be bound by all applicable laws, acts, statutes, rules, regulations, ordinances, codes, privileges, treaties, agreements, directives, provisions, injunctions, declarations, ordinances, decrees, or other laws of any kind whatsoever of any applicable governmental body, authority, agency, court, committee, or any applicable judicial body.

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Award Mechanism

In order to ensure clarity and transparency in the procurement process, it is important to determine the mechanism by which contracts are awarded at the end of the competition process, to this end, this Request for Qualifications (RFQ) serves as the initial stage of a two-tier selection process. During the evaluation of qualification submissions, bidders who meet and exceed the established qualification criteria for the subsequent stage—the Request for Proposals (RFP)—will be notified. The RFP stage provides an opportunity for the selected bidders to submit additional details, and serves as a more in-depth exploration of each bidder's capabilities and proposed solutions, thereby enabling a comprehensive assessment to ensure the selection of the most suitable candidate for contract award. Accordingly, participation in the RFP stage is contingent upon successfully passing the qualification requirements during this RFQ stage..

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Section Five - Qualification Applications Forms

Applicant Information Form

Advanced			
	Value of the applicant's capital		Name of the applicant (company/institution)
	Date		Commercial Registration Number
	Function		Name of the person responsible
Address			
	Street		City
	Zip code		P.O. Box
	Fax		Phone
	Establishment Date		Email
			Site
Company Ownership			
Partnership Percentage	Nationality	Owners/Partners	
Company Representative Information			
		Name	
		Job Title	
Mobile:	Telephone	Phone	
		Email	

Figure 7: Applicant Information Form

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Technical and administrative capabilities

Technical and Administrative Capabilities	
Previous Work Experience	
Number of years of experience in the field of project	
Number of similar projects executed in the last three years	
Average results of performance evaluations in previous projects with government agencies in the field of project	
Quality	
What are the quality assurance standards? (provide evidence)	
Environment, Health and Safety	
What are the quality assurance standards for environmental, health and safety?	
Existing contractual liabilities	
Number of existing projects	
Value of existing projects	
Human Resources	
Number of employees	
Number of Saudi employees	
Percentage of Saudi employees	

Figure 8: Technical and managerial capacities

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Administrative Staff Experience Model

Describe the experiences of the applicant's administrative staff as follows:				
Number of years of experience	Specialization / Area of Expertise	Function	Name	#
				١
				٢
				٣
				٤
				٥
				٦
				٧
				٨
				٩
				10

Figure 9: Experience of administrative staff

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Professional Staff Experience Model

The expertise of the applicant's technical staff is as follows:				
Number of years of experience	Specialization/ Area of Expertise	Function	Name	#
				١
				٢
				٣
				٤
				٥
				٦
				٧
				٨
				٩
				١٠
				١١
				١٢
				١٣
				١٤
				١٥
				١٦
				١٧
				١٨
				١٩
				20

Figure 10: Professional Staff Experience

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Experience in similar projects

Write details of three similar projects that have been implemented over the past five years as follows:		
The first project		soap opera
	Project Name	1
	Project Location	2
	Project Content	3
	Project Owner	4
	Contract Value	5
	Contract Duration	6
	Start Date	7
	Delivery Date	8
	Name of the person responsible	9
	Contact information for the person in charge	10
	Email of the person in charge	11
The second project		soap opera
	Project Name	1
	Project Location	2
	Project Content	3
	Project Owner	4
	Contract Value	
	Contract Duration	6
	Start Date	7
	Delivery Date	8
	Name of the person responsible	9
	Contact information for the person in charge	10
	Email of the person in charge	11
The third project		soap opera
	Project Name	1
	Project Location	2
	Project Content	3
	Project Owner	4
	Contract Value	5
	Contract Duration	6

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	Start Date	7
	Delivery Date	8
	Name of the person responsible	9
	Contact information for the person in charge	10
	Email of the person in charge	11

Figure 11: Experience in similar projects

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Financial Capacity Criteria

القدرات المالية			
البيانات المالية			
-2	-1	0	البيانات المالية لآخر 3 سنوات مالية
بيان الميزانية العمومية			
			الأصول المتداولة
			النقدية ومكافآت النقدية
			الحسابات مستحقة القبض
			الالتزامات المتداولة
مؤشرات الأداء المالية			
			نسبة النقدية (النقدية ومكافآت النقدية) / (الالتزامات المتداولة)
			نسبة التداول (الأصول المتداولة) / (الالتزامات المتداولة)
			نسبة السيولة السريعة ((النقدية ومكافآت النقدية + الحسابات المستحقة القبض) / (الالتزامات المتداولة))
المرفقات			
البيانات المالية لآخر 3 سنوات مالية للمتقدم بالتأهيل (ليس للشركة الأم أو الشركات التابعة)			

Shape 124: criteria Capacity Finance

Note: Forms must be filled out based on what is required on the website (Etimad platform)

Request for Qualification for the Proposal of the Geological Data Analysis Center Platform

Abbreviations

The following standard abbreviations are used in the Terms and Conditions booklet, and other abbreviations are defined when they are mentioned in the manual.

Explain	Icon
Saudi Center for Geological Data Analysis	GDAC-SA
Level of Application Services	ASL
Expenditure Efficiency and Government Projects Authority	EXPRO
General Authority for Survey and Geospatial Information	GEOSA
GIS	GIS
Instructions for Bidders	ITB
Saudi Arabia	KSA
Switch Label Protocols	MPLS
National Drilling Library	NCL
National Geological Database	NGD
National Information Center	NIC
Quality Assurance / Quality Control	QA/QC
Terms & Specifications Brochure	RFP
General Program of the Geological Survey	RGP
Research and Training Institute	RTI
Saudi National Reference System	SANSRS
Department of Scientific Data Analysis	SDAD
Saudi Data and Artificial Intelligence Authority	SDAIA
Saudi Geological Survey	SGS
Virtual Private Network	VPN

Shape ∴ abbreviations